Complement-packed coefficients for a Diophantine presentation of ZFC consistency

OpenAI-assisted research for Vladimir Reshetnikov and ProveIt

23 October 2026

Pinned comparison: ProveIt commit `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`, package 36.

1 Complement-packed native clauses

This section gives an independent structural reduction of the complete coefficient tuple at the inherited 67-operation polynomial and 59-operation system frontiers. Their arithmetic circuits, degrees, and positive-variable domains do not change. The old tuple in package 36 had $B = 2^{1012115}$ and occupied 506664 raw bytes. The new tuple has

$$\rho = 2^{67}, \qquad L = 8807, \qquad d = 590069, \qquad B = \rho^L = 2^{590069},$$

and its four materialized ports occupy 295388 raw bytes, a reduction of 41.6994%. This changes the integers themselves, rather than merely their file compression.

1.1 One congruence replaces the parity fields

Lemma 1 (Aggregate complement decoder)**.** *Let $N < 2^a$, let $0 < v_i < 2^m$ for $0 \le i < N$, and let all x_i, y_j be Boolean. Define*

$$n = \sum_i x_i, \quad W = \sum_{j=0}^{m-1} 2^j y_j, \quad S = \sum_i \big(1 + 2^a(2^m - v_i)\big)x_i + 2^a W, \quad \mu = 2^{a+m} - 2.$$

Then $S \,\mathrm{AND}\, \mu = 0$ if and only if either all selectors and copies vanish, or exactly one selector i is present and $W = v_i$.

Proof. The low a bits of S equal n, since $0 \le n \le N < 2^a$. Their mask is $2^a - 2$, so $n \in \{0, 1\}$. This extraction precedes, and does not assume, any claim about internal payload carries. If $n = 0$, the remaining payload is $W < 2^m$, whose low m bits must vanish, so $W = 0$. If selector i is present, the payload is $2^m - v_i + W \in (0, 2^{m+1})$. Its low m bits vanish exactly when it equals 2^m, hence exactly when $W = v_i$. Conversely valid cells have $S = 0$ or $S = 2^{a+m} + 1$, both disjoint from μ. □

The interval $W < 2^m$ uses one distinct Boolean support position for each copy bit. It does not apply to unconstrained nonnegative copy values. The native low mask supplies these Boolean conditions; both dummy positions have clause coefficient zero.

Retain the exact Wang943 language and its ordering. For a tile with west, north, east, south labels (w, n, e, s), put

$$v_i = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

The lower 34 bits encode the labels. Positions $34, 35, 36, 37$ are the four occupancy bits, and positions $38, 39, 40$ are the three anchors. These last seven bits are one for every occupied tile. Use $a = 10, m = 41$, so the selector coefficient and mask are

$$c_i = 1 + 1024(2^{41} - v_i), \qquad \mu = 2^{51} - 2.$$

Moving the seven constant bits to the top gives $c_i \le 1 + 2^{44}$. This additionally improves the uniform carry bound.

The 38 native edge-copy positions are unchanged. Their payload positions are specified by

slot	native positions	payload bit positions
W	$1023 + j,\ 0 \le j < 9$	$0, \dots, 7, 34$
N	$1013 + j,\ 0 \le j < 10$	$8, \dots, 16, 35$
E	$1003 + j,\ 0 \le j < 9$	$17, \dots, 24, 36$
S	$983 + j,\ 0 \le j < 10$	$25, \dots, 33, 37$

The anchors $1095, 2195, 3296$ carry payload positions $38, 39, 40$. Let E be these 41 positions together with selectors $0, \dots, 942$ and dummies $1063, 1064$; thus $|E| = 986$. At the copy position for bit j put $c_e = 2^{10+j}$, and at each dummy put $c_e = 0$. The lemma now proves exactly the same local valid cells as the old 41 separate radix-four parity fields.

1.2 The full coefficient recipe

All powers below have fixed finite exponents. Define

$$\begin{aligned}
D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e \in E} c_e(\rho^{4393-e} + \rho^{5495-e}),\\
D_R &= \rho^{42},\\
Z &= (2^{65} - 1) \sum_{e=8792}^{8804} \rho^e + (2^{19} - 1)\rho^{8805},\\
M_C &= B - 1 - \sum_{e \in E \setminus \{1\}} \rho^e - 2\rho^{1063},\\
M_{F,\text{native}} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{8} \rho^{1065+j} + \rho^{2195} + \rho^{3296} + Z.
\end{aligned}$$

The four arithmetic ports, all of which must be replaced together, are

$$\boxed{\beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_{F,\text{native}} + B - 1.}$$

The fourth port is shifted; substituting the native high mask is incorrect.

The old support geometry is preserved. The two clause targets are $4393, 5495$; the two distinct tested temporal anchors remain $2195, 3296$ with shared current source 1095. Finite support arithmetic verifies

$$((4393 - E) \bmod L) \cap ((5495 - E) \bmod L) = \varnothing, \qquad ((2195 - E) \bmod L) \cap ((3296 - E) \bmod L) = \{0\}.$$

All 66062 current/spatial/temporal native-basis identities are verified independently. The aggregate lemma at the clean band of the distinguished Start cell forces its shared anchor, and hence both tested temporal anchors, to one. The inherited rotation argument therefore still excludes binary misalignment and then nonzero whole-radix cell misalignment. After alignment, overlap parity recovers the original occupied edge codes and the same Wang matching constraints.

For the inherited conservative uniform bound one now has

$$\mathcal{M} = 988\left(2\sum_{e\in E} c_e + 6\right) = 29613572795083482608, \qquad 4\mathcal{M}+8 = 118454291180333930440 < 2^{67}.$$

Also $2\mu + 4 < 2^{67}$ and $2(2^{65}-1)+4 < 2^{67}$. These bounds preserve the compiler's carry, positive-slack, and nonaligned-rotation estimates. The new internal payload carries are intentional and are covered by the lemma, not by an assumption of separate fields.

The largest unshifted product exponent is 8791. Each padded test, at a position from 8792 through 8805, therefore constrains a zero in an aligned cell. There are 864 such tests. Consequently

$$\mathrm{pc}(M_C) = d - 986, \quad \mathrm{pc}(M_{F,\mathrm{native}}) = 986, \quad M_C \equiv 2 \pmod 4, \quad M_{F,\mathrm{native}} \equiv 4 \pmod 8.$$

The original native mask population balance and low congruences remain exact.

1.3 The new actual-rank controller

For completeness use the explicitly certified prime

$$\lambda = 581197, \qquad k = \lambda - 1 = 581196, \qquad d = 67 \cdot 8807.$$

The multiplicative orders of B modulo $67, 8807, \lambda$ are respectively $66, 4403, 581196$. Their required divisibility conditions hold, and

$$B^k \bmod \lambda^2 = 182964883980 \neq 1, \qquad B^k \equiv 1 \pmod \lambda.$$

For $H_0 = 1 + BD_C + B^2(D_R + 1)$, the residues

p	B	B^2-1	H_0
67	46	38	41
8807	456	5374	5193
581197	348469	573553	76889

prove the necessary unit conditions. Choose sufficiently large independent spatial and temporal dimensions w, h that are powers of λ, and put $N = wh$, $q = B^N$. Switching a safe low dummy bit at position j changes the actual packed rank by

$$\delta R = -\Gamma B^j, \qquad \Gamma = \rho^{1063}(q^2-1)[1 + q(D_C + BD_R + B^w)].$$

The certificates imply $\gcd(\Gamma, dN) = 1$. With $T = N/\lambda$, the powers B^{kj} run through all residues $1 + d\lambda z$ modulo dN. The original Boolean consecutive-orbit subset argument, using $\{B^{2+kj} : 0 \le j < T\} \cup \{B^3\}$, therefore adjusts the actual rank to $R \equiv dw \pmod{dN}$. Taking $N > d\lambda^2$ and $h > \lambda$ keeps all chosen dummy cells and their needed translates below N and away from the origin. The complete argument is reproduced in `native/PROOF.md`; it is not a test of only synthetic tableaux.

Both d and these canonical dimensions are odd. The unchanged population and congruence identities retain the source square/odd-scale promises. Soundness, unlike completeness, places no prime-power restriction on the dimensions of an arbitrary recovered tableau.

1.4 The short explicit system

Use sixteen strictly positive unknowns

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma.$$

Define the following polynomial abbreviations, not additional unknowns:

$$q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \quad Y = sq^3,$$
$$E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y,$$
$$H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H,$$
$$G = (q-1)(q-F) + \alpha, \quad R = G(q^2-1) + (M_C + qM_F)J,$$
$$L_0 = EkY, \quad V = of - c, \quad U = jc - R, \quad I = ic^2,$$
$$A_0 = \Delta(f^2 - 1).$$

The seven equations are

$$\tau^2 - L_0(L_0 + k) = 1, \qquad D^2 - \Delta c^2 = 1,$$
$$A_0(V^2 - y^2) + y^2 = 1, \qquad k - hE - R = 1,$$
$$(K_0 + w)C = F + z(q-1), \qquad I^2 = A_0, \qquad U = V.$$

The exact evaluator remains 59 operations. By the inherited arithmetic theorem, the native interface proved here, the unchanged Wang language, and the literal K215 calibration, its positive unsolvability is provably equivalent in ZFC to Con(ZFC). The supplied 67-operation polynomial is rebound by the same complete tuple and keeps its fifteen positive unknowns and exact degree 155.

1.5 Reproduction and limits

The builder reconstructs the literal tile language from all 430 pinned machine actions without executing the machine. The independent auditor starts instead from the pinned tile list, imports no builder code, reconstructs all four full integers, verifies the source Git blob identities, and checks all native basis identities, exact carry and mask bounds, and the prime controller. It also verifies that both rebound arithmetic sources are unchanged. The complement lemma and inherited universal compiler proofs, not finite diagnostic examples alone, supply the unbounded mathematical conclusion. No new end-to-end proof-assistant certificate, consistency decision, or global optimum is claimed.